

Weimer et al. (2023): A Causal Approach for Detecting Team-Level Momentum in NBA Games

1. Research Question

Does **team-level momentum** exist in basketball?

Momentum hypothesis:

$$P(\text{score next possession} \mid \text{recent scoring run}) > P(\text{score next possession})$$

Traditional skepticism:

- Many studies treat scoring as independent events.
- Momentum often attributed to cognitive bias (“hot hand illusion”).

This paper tests whether **timeouts interrupt momentum**. If pauses reduce scoring by the momentum team, momentum must exist.

Key idea:

If momentum does not exist $\Rightarrow$ timeouts should not affect scoring.

2. Causal Identification Strategy

Problem:

- Coaches call timeouts strategically.
- Strategic timeouts introduce **selection bias**.

Solution: Use **TV timeouts**.

TV timeouts are:

exogenous stoppages of play

They occur at predetermined times in the NBA rulebook and are unrelated to momentum.

Thus:

TV timeout $\approx$ random assignment of pause in play

This approximates a randomized experiment.

3. Matching Framework

Goal: Compare runs that are identical except for the presence of a TV timeout.

Matching variables:

- run length
- substitutions by momentum team
- substitutions by opposing team

Propensity score matching:

$$Pr(Treatment_i = 1 \mid X_i)$$

where

$$X_i = (\text{run size, substitutions})$$

Each treated run (interrupted by TV timeout) is matched with a similar untreated run.

4. Operational Definition of Momentum

Momentum indicator:

$$m_i = \begin{cases} 1 & \text{if } p_u \geq 6 \\ 0 & \text{otherwise} \end{cases}$$

where

$$p_u = \text{unanswered points.}$$

Thus momentum begins when:

$$\text{run} \geq 6 \text{ points.}$$

Robustness checks also test:

$$10 \text{ point runs, } 15 \text{ point runs.}$$

Outcome variable:

$$Y_i = \text{points scored by the momentum team in the next 3 minutes.}$$

5. Statistical Model

Outcome modeled with Poisson regression:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{TVTimeout}_i + \beta_2 \text{RunSize}_i + \beta_3 \text{SubMomentum}_i + \beta_4 \text{SubOpponent}_i$$

Key coefficient:

$$\beta_1 < 0$$

indicates that timeouts reduce scoring during momentum.

6. Main Result

TV timeouts reduce scoring by the momentum team:

$$e^{-0.119} - 1 \approx -11.2\%$$

Interpretation:

Momentum team scores $\approx 11\%$ fewer points after TV timeout.

Average effect:

$$\approx 0.56 \text{ fewer points over the next 3 minutes.}$$

Stronger effects appear for longer runs.

7. Mechanisms

Two mechanisms explain the effect:

1. Pause in play

Stopping gameplay interrupts momentum.

2. Strategy changes

Substitutions or tactical adjustments.

Evidence:

- Timeouts with no substitutions still reduce scoring.
- Strategy explains only about 40% of the effect.

Thus most of the effect comes from the **stoppage itself**.

8. Robustness Checks

Several tests confirm results:

- different run thresholds (6, 10, 15 points)
- alternative time windows for scoring
- negative binomial models

- Monte Carlo simulation with no momentum

Simulation result:

If momentum = 0 $\Rightarrow$ timeouts have no effect.

Thus the empirical effect is not a statistical artifact.

9. Interpretation

Momentum behaves like a **state variable**:

$$\text{performance}_{t+1} = f(\text{performance}_t)$$

Pausing play disrupts that state.

This implies:

- momentum exists at the **team level**
- interruptions reduce performance

10. Implications

Strategic insights:

- Timeouts help stop opponent momentum.
- Momentum teams should avoid substitutions.

Methodological contribution:

- Demonstrates how **natural experiments** can detect causal effects in sports.
- Shows value of **matching methods in observational data**.

Key Insight

Momentum in basketball is not purely psychological.

Performance is serially correlated at the team level.

Interrupting play breaks that correlation.